Picard rank jumps for non-ordinary K3 surfaces

(joint work with Ruofan Jiang and Ananth Shankar)

Ziquan Yang

CUHK/IMS

May 1st, 2026

A Motivating Question

- Let's say we have a surface $X \subseteq \mathbb{P}^3$, given by a homogenous polynomial

$$f(X_0, X_1, X_2, X_3) = 0.$$

A Motivating Question

- Let's say we have a surface $X \subseteq \mathbb{P}^3$, given by a homogenous polynomial

$$f(X_0, X_1, X_2, X_3) = 0.$$

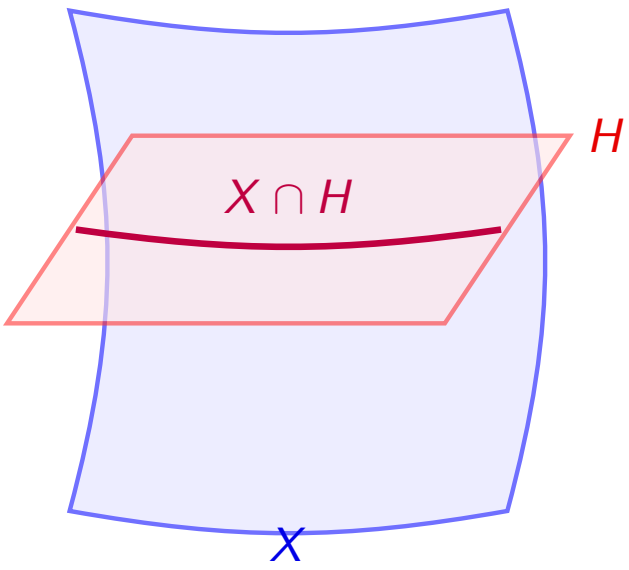
- How to construct curves on X?

A Motivating Question

- Let's say we have a surface $X \subseteq \mathbb{P}^3$, given by a homogenous polynomial

$$f(X_0, X_1, X_2, X_3) = 0.$$

- How to construct curves on X?
- Naively, take another surface $H \subseteq \mathbb{P}^3$. Then take $X \cap H$.

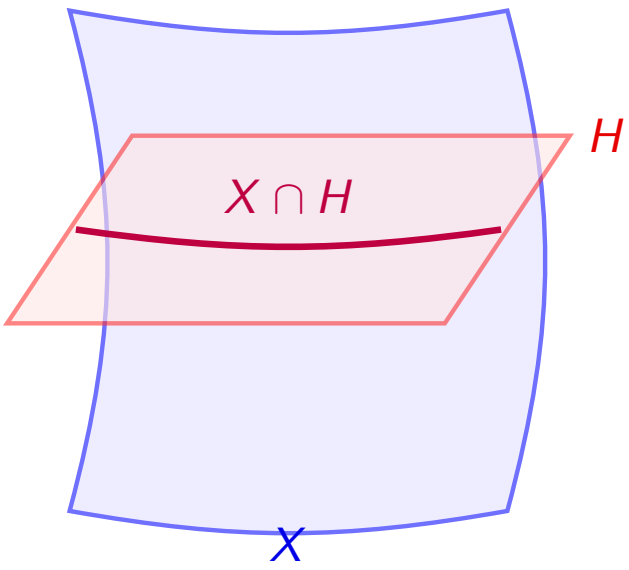


A Motivating Question

- Let's say we have a surface $X \subseteq \mathbb{P}^3$, given by a homogenous polynomial

$$f(X_0, X_1, X_2, X_3) = 0.$$

- How to construct curves on X?
- Naively, take another surface $H \subseteq \mathbb{P}^3$. Then take $X \cap H$.
- The resulting curve is by definition a complete intersection curve.



Origins of Noether-Lefschetz Theory

Theorem (Noether-Lefschetz)

For a very general surface $X \subseteq \mathbb{P}^3_{\mathbb{C}}$ of degree $d \geq 4$, the restriction map

$$\mathsf{Pic}(\mathbb{P}^3_{\mathbb{C}}) \to \mathsf{Pic}(X) \text{ given by } [H] \mapsto [X \cap H]$$

is an isomorphism.

Origins of Noether-Lefschetz Theory

Theorem (Noether-Lefschetz)

For a very general surface $X \subseteq \mathbb{P}^3_{\mathbb{C}}$ of degree $d \geq 4$, the restriction map

$$\mathsf{Pic}(\mathbb{P}^3_{\mathbb{C}}) \to \mathsf{Pic}(X) \text{ given by } [H] \mapsto [X \cap H]$$

is an isomorphism.

- This means that for a "very general" surface of degree 4 or higher, every curve on the surface is simply a complete intersection with another surface.

Origins of Noether-Lefschetz Theory

Theorem (Noether-Lefschetz)

For a very general surface $X \subseteq \mathbb{P}^3_{\mathbb{C}}$ of degree $d \geq 4$, the restriction map

$$\mathrm{Pic}(\mathbb{P}^3_{\mathbb{C}}) \to \mathrm{Pic}(X) \text{ given by } [H] \mapsto [X \cap H]$$

is an isomorphism.

- This means that for a "very general" surface of degree 4 or higher, every curve on the surface is simply a complete intersection with another surface.
- However, there are exceptions. The parameter space of surfaces containing non-complete intersection curves forms a union of proper subvarieties.

Origins of Noether-Lefschetz Theory

Theorem (Noether-Lefschetz)

For a very general surface $X \subseteq \mathbb{P}^3_{\mathbb{C}}$ of degree $d \geq 4$, the restriction map

$$\mathrm{Pic}(\mathbb{P}^3_{\mathbb{C}}) \to \mathrm{Pic}(X) \text{ given by } [H] \mapsto [X \cap H]$$

is an isomorphism.

- This means that for a "very general" surface of degree 4 or higher, every curve on the surface is simply a complete intersection with another surface.
- However, there are exceptions. The parameter space of surfaces containing non-complete intersection curves forms a union of proper subvarieties.
- The locus of those surfaces with more curves is known as the Noether-Lefschetz (NL) locus.

An Explicit Example: The Fermat Quartic

- Consider the Fermat quartic surface $X \subseteq \mathbb{P}^3_{\mathbb{C}}$ defined by:

$$X_0^4 + X_1^4 + X_2^4 + X_3^4 = 0$$

An Explicit Example: The Fermat Quartic

- Consider the Fermat quartic surface $X \subseteq \mathbb{P}^3_{\mathbb{C}}$ defined by:

$$X_0^4 + X_1^4 + X_2^4 + X_3^4 = 0$$

- The line L given by the equations $X_0 = \zeta X_1$ and $X_2 = \zeta X_3$ (where $\zeta^4 = -1$) lies entirely on X.

An Explicit Example: The Fermat Quartic

- Consider the Fermat quartic surface $X \subseteq \mathbb{P}^3_{\mathbb{C}}$ defined by:

$$X_0^4 + X_1^4 + X_2^4 + X_3^4 = 0$$

- The line L given by the equations $X_0 = \zeta X_1$ and $X_2 = \zeta X_3$ (where $\zeta^4 = -1$) lies entirely on X.
- By Bézout's theorem, any complete intersection curve on a quartic surface must have a degree that is a multiple of 4.

An Explicit Example: The Fermat Quartic

- Consider the Fermat quartic surface $X \subseteq \mathbb{P}^3_{\mathbb{C}}$ defined by:

$$X_0^4 + X_1^4 + X_2^4 + X_3^4 = 0$$

- The line L given by the equations $X_0 = \zeta X_1$ and $X_2 = \zeta X_3$ (where $\zeta^4 = -1$) lies entirely on X.
- By Bézout's theorem, any complete intersection curve on a quartic surface must have a degree that is a multiple of 4.
- Since $\deg(L) = 1$, the line L cannot be a complete intersection, meaning $\operatorname{rank} \operatorname{Pic}(X) > 1$.

An Explicit Example: The Fermat Quartic

- Consider the Fermat quartic surface $X \subseteq \mathbb{P}^3_{\mathbb{C}}$ defined by:

$$X_0^4 + X_1^4 + X_2^4 + X_3^4 = 0$$

- The line L given by the equations $X_0 = \zeta X_1$ and $X_2 = \zeta X_3$ (where $\zeta^4 = -1$) lies entirely on X.
- By Bézout's theorem, any complete intersection curve on a quartic surface must have a degree that is a multiple of 4.
- Since $\deg(L) = 1$, the line L cannot be a complete intersection, meaning $\operatorname{rank} \operatorname{Pic}(X) > 1$.
- Given a smooth proper family $f : X \to S$, the NL locus is where the (geometric) fibers acquire more divisors (or line bundles) than the (geometric) generic fiber (i.e., the Néron-Severi group "jumps").

The Role of Hodge Theory

- Question: Can "very general" be replaced by "general" in the Noether-Lefschetz theorem?

The Role of Hodge Theory

- Question: Can "very general" be replaced by "general" in the Noether-Lefschetz theorem?
- The answer is NO (Ciliberto-Harris-Miranda). Over $\mathbb{C}$, this can be seen quite explicitly using classical Hodge theory.

The Role of Hodge Theory

- Question: Can "very general" be replaced by "general" in the Noether-Lefschetz theorem?
- The answer is NO (Ciliberto-Harris-Miranda). Over $\mathbb{C}$, this can be seen quite explicitly using classical Hodge theory.

Lefschetz (1,1)-Theorem

Let X be a smooth projective variety over $\mathbb{C}$. Then

$$\mathrm{NS}(X) \simeq \mathrm{H}^2(X, \mathbb{Z}) \cap \mathrm{H}^1(X, \Omega_X) \subseteq \mathrm{H}^2_{\mathrm{dR}}(X/\mathbb{C}).$$

The Role of Hodge Theory

- Question: Can "very general" be replaced by "general" in the Noether-Lefschetz theorem?
- The answer is NO (Ciliberto-Harris-Miranda). Over $\mathbb{C}$, this can be seen quite explicitly using classical Hodge theory.

Lefschetz (1,1)-Theorem

Let X be a smooth projective variety over $\mathbb{C}$. Then

$$\mathrm{NS}(X) \simeq \mathrm{H}^2(X, \mathbb{Z}) \cap \mathrm{H}^1(X, \Omega_X) \subseteq \mathrm{H}^2_{\mathrm{dR}}(X/\mathbb{C}).$$

- As X varies in a family, the integral lattice $\mathrm{H}^2(X, \mathbb{Z})$ remains rigid locally, but the complex subspace $\mathrm{H}^{1,1}(X)$ varies continuously.

The Role of Hodge Theory

- Question: Can "very general" be replaced by "general" in the Noether-Lefschetz theorem?
- The answer is NO (Ciliberto-Harris-Miranda). Over $\mathbb{C}$, this can be seen quite explicitly using classical Hodge theory.

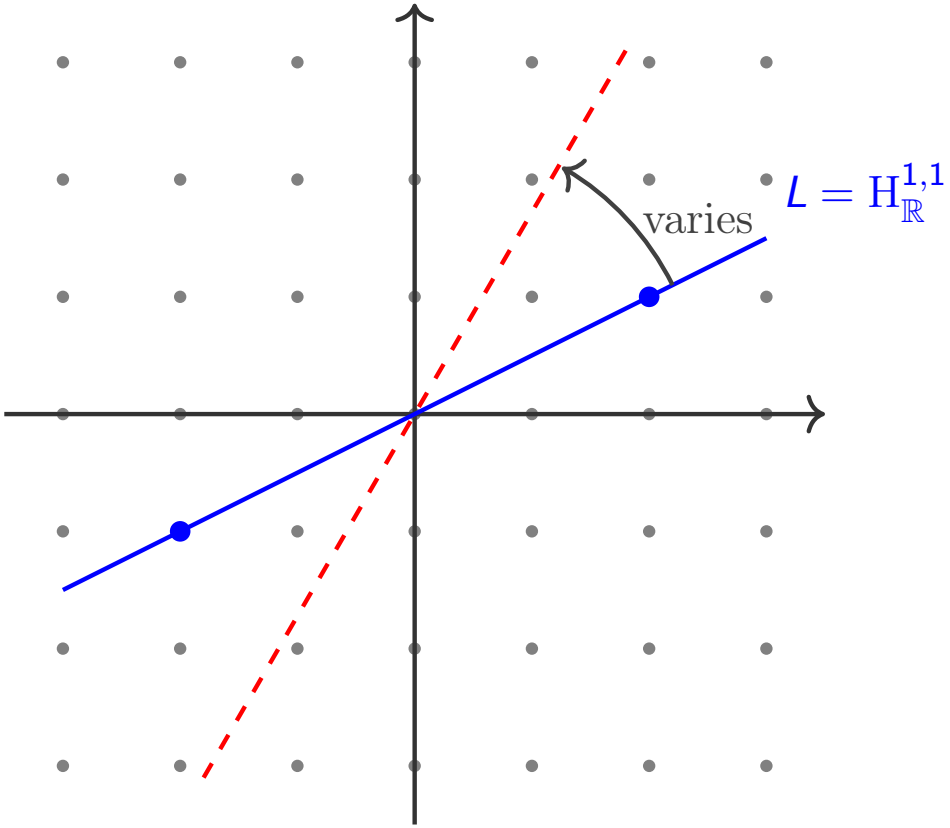
Lefschetz (1,1)-Theorem

Let X be a smooth projective variety over $\mathbb{C}$. Then

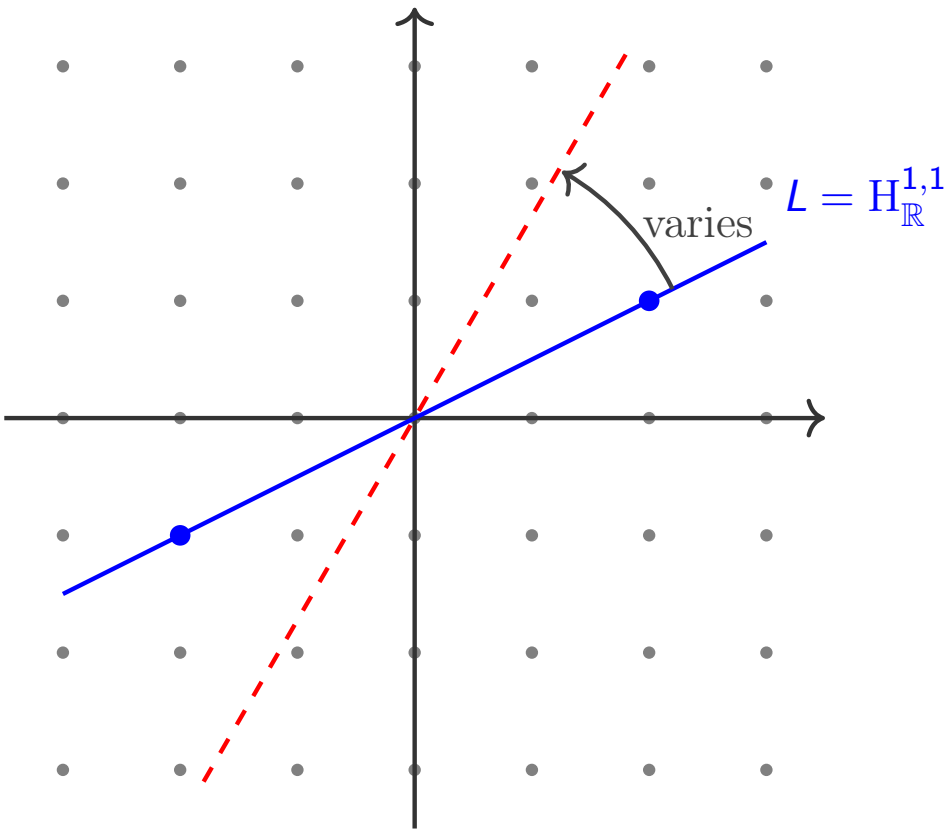
$$\mathrm{NS}(X) \simeq \mathrm{H}^2(X, \mathbb{Z}) \cap \mathrm{H}^1(X, \Omega_X) \subseteq \mathrm{H}^2_{\mathrm{dR}}(X/\mathbb{C}).$$

- As X varies in a family, the integral lattice $\mathrm{H}^2(X, \mathbb{Z})$ remains rigid locally, but the complex subspace $\mathrm{H}^{1,1}(X)$ varies continuously.
- The NL locus is exactly the dense, countable union of subvarieties where extra integral classes unexpectedly align with $\mathrm{H}^{1,1}(X)$.

A Cartoon for the NL Locus

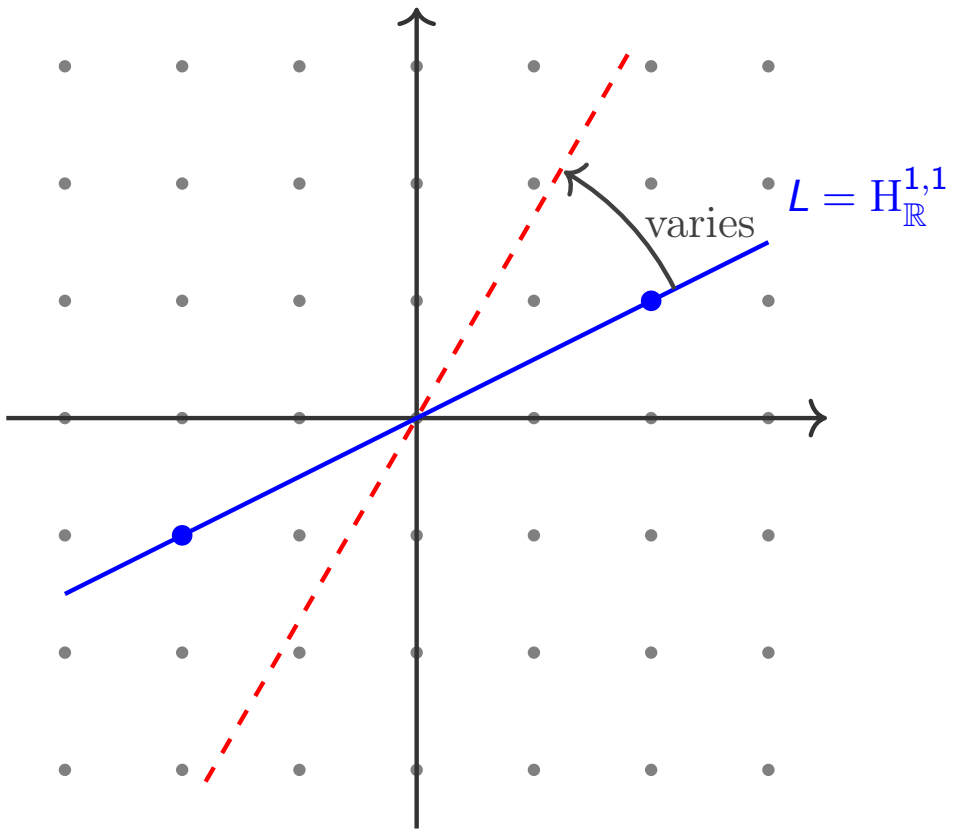


A Cartoon for the NL Locus



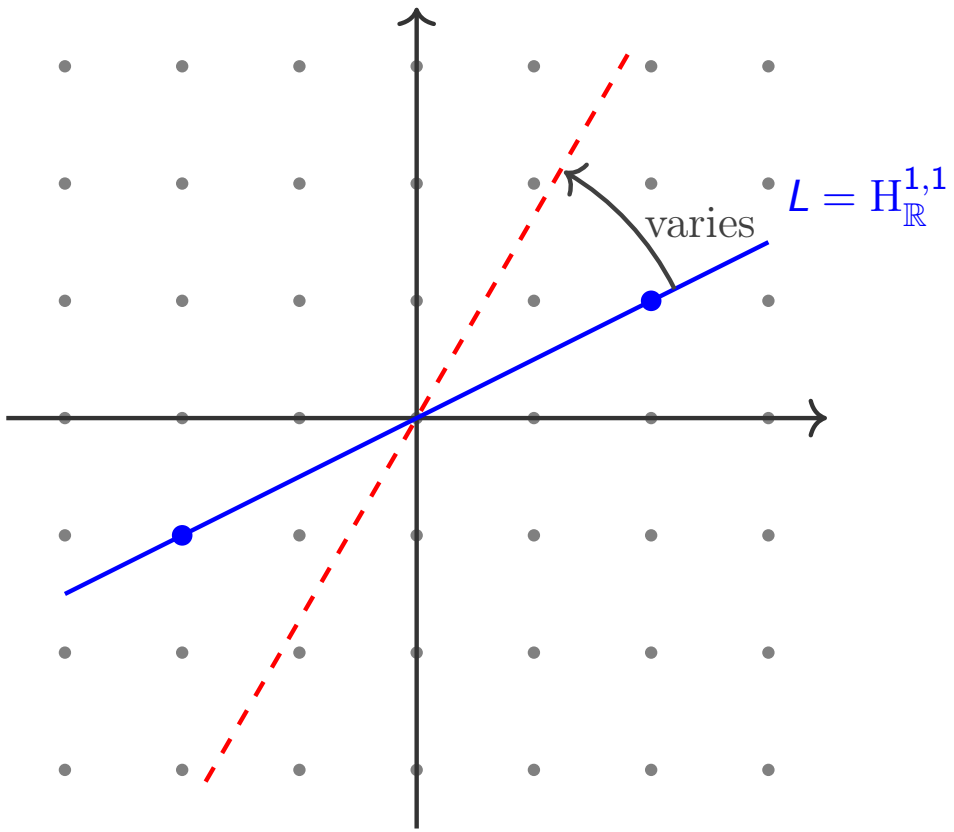
- We can picture $\mathrm{H}^2(X, \mathbb{R})$ as the plane $\mathbb{R}^2$, and the integral cohomology $\mathrm{H}^2(X, \mathbb{Z})$ as the lattice $\mathbb{Z}^2$.

A Cartoon for the NL Locus



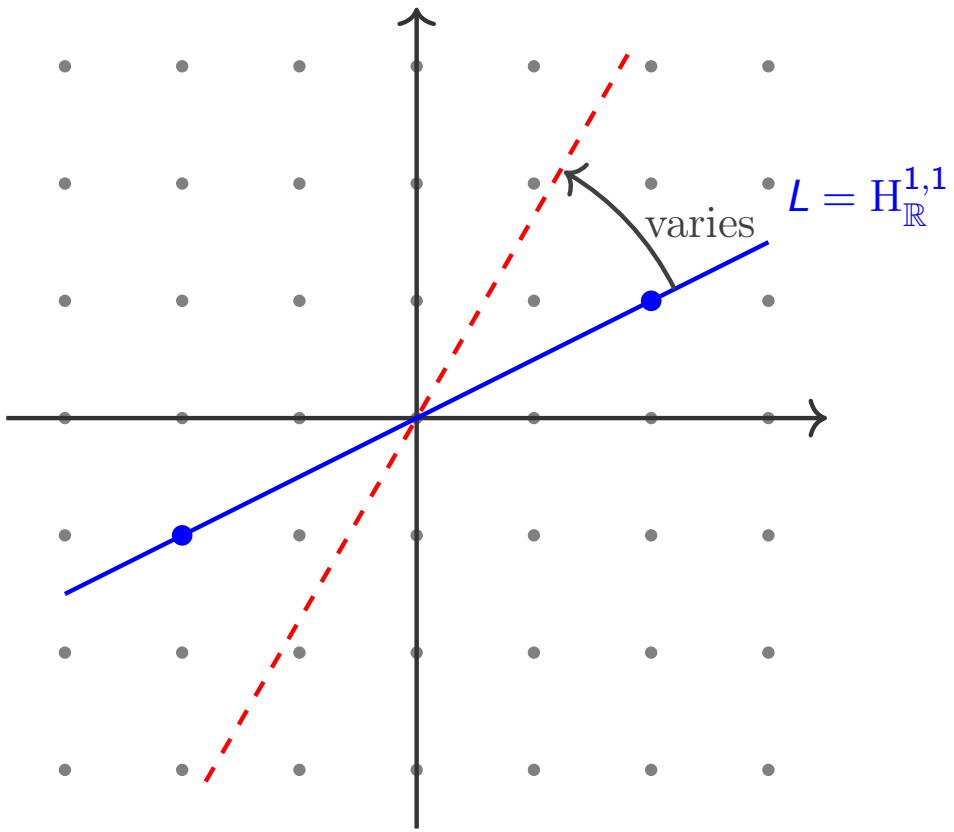
- We can picture $\mathrm{H}^2(X, \mathbb{R})$ as the plane $\mathbb{R}^2$, and the integral cohomology $\mathrm{H}^2(X, \mathbb{Z})$ as the lattice $\mathbb{Z}^2$.
- Say $\mathrm{H}^{1,1}(X) \cap \mathrm{H}^2(X, \mathbb{R})$ is represented by a line L passing through the origin.

A Cartoon for the NL Locus



- We can picture $\mathrm{H}^2(X, \mathbb{R})$ as the plane $\mathbb{R}^2$, and the integral cohomology $\mathrm{H}^2(X, \mathbb{Z})$ as the lattice $\mathbb{Z}^2$.
- Say $\mathrm{H}^{1,1}(X) \cap \mathrm{H}^2(X, \mathbb{R})$ is represented by a line L passing through the origin.
- As the surface X varies in our family, the line L rotates continuously.

A Cartoon for the NL Locus



- We can picture $\mathrm{H}^2(X, \mathbb{R})$ as the plane $\mathbb{R}^2$, and the integral cohomology $\mathrm{H}^2(X, \mathbb{Z})$ as the lattice $\mathbb{Z}^2$.
- Say $\mathrm{H}^{1,1}(X) \cap \mathrm{H}^2(X, \mathbb{R})$ is represented by a line L passing through the origin.
- As the surface X varies in our family, the line L rotates continuously.
- L intersects the integral lattice (yielding new divisors) exactly when its gradient is a rational number.

Green's Theorem

- This cartoon can be made precise by considering the variation of Hodge structures on $\mathbb{V} := R^2 f_* \mathbb{Q}$.

Green's Theorem

- This cartoon can be made precise by considering the variation of Hodge structures on $\mathbb{V} := R^2 f_* \mathbb{Q}$.
- Infinitesimally, the variation of Hodge structures is captured by the Kodaira-Spencer map:

$$\nabla_s \colon T_s S \to \mathrm{Hom}(\mathbb{V}^{1,1}_{dR,s}, \mathbb{V}^{0,2}_{dR,s})$$

Green's Theorem

- This cartoon can be made precise by considering the variation of Hodge structures on $\mathbb{V} := R^2 f_* \mathbb{Q}$.
- Infinitesimally, the variation of Hodge structures is captured by the Kodaira-Spencer map:

$$\nabla_s \colon T_s S \to \mathrm{Hom}(\mathbb{V}^{1,1}_{dR,s}, \mathbb{V}^{0,2}_{dR,s})$$

Theorem (Green)

If for some $\lambda \in \mathbb{V}^{1,1}_{dR,s}$ the map $\nabla_s \lambda \colon T_s S \to \mathbb{V}^{0,2}_{dR,s}$ is surjective, then the NL locus is analytically dense.

Green's Theorem

- This cartoon can be made precise by considering the variation of Hodge structures on $\mathbb{V} := R^2 f_* \mathbb{Q}$.
- Infinitesimally, the variation of Hodge structures is captured by the Kodaira-Spencer map:

$$\nabla_s \colon T_s S \to \mathrm{Hom}(\mathbb{V}^{1,1}_{dR,s}, \mathbb{V}^{0,2}_{dR,s})$$

Theorem (Green)

If for some $\lambda \in \mathbb{V}^{1,1}_{dR,s}$ the map $\nabla_s \lambda \colon T_s S \to \mathbb{V}^{0,2}_{dR,s}$ is surjective, then the NL locus is analytically dense.

- Over $\mathbb{C}$, the variation for surfaces in $\mathbb{P}^3$ is rich enough, so that the NL locus for surfaces in $\mathbb{P}^3$ is analytically dense.

The Case of K3 Surfaces

- When $h^{0,2}(\mathbb{V}) = 1$ (e.g., for K3 surfaces), the hypothesis of Green's theorem just amounts to saying that $\mathbb{V}$ is non-isotrivial.

The Case of K3 Surfaces

- When $h^{0,2}(\mathbb{V}) = 1$ (e.g., for K3 surfaces), the hypothesis of Green's theorem just amounts to saying that $\mathbb{V}$ is non-isotrivial.
- This special case was also observed by Oguiso.

The Case of K3 Surfaces

- When $h^{0,2}(\mathbb{V}) = 1$ (e.g., for K3 surfaces), the hypothesis of Green's theorem just amounts to saying that $\mathbb{V}$ is non-isotrivial.
- This special case was also observed by Oguiso.

Question

What happens when we replace the base S by an arithmetic curve?

The Case of K3 Surfaces

- When $h^{0,2}(\mathbb{V}) = 1$ (e.g., for K3 surfaces), the hypothesis of Green's theorem just amounts to saying that $\mathbb{V}$ is non-isotrivial.
- This special case was also observed by Oguiso.

Question

What happens when we replace the base S by an arithmetic curve?

- Say $S = \mathrm{Spec}\,(\mathcal{O}_K)$ for some number field K.

The Case of K3 Surfaces

- When $h^{0,2}(\mathbb{V}) = 1$ (e.g., for K3 surfaces), the hypothesis of Green's theorem just amounts to saying that $\mathbb{V}$ is non-isotrivial.
- This special case was also observed by Oguiso.

Question

What happens when we replace the base S by an arithmetic curve?

- Say $S = \mathrm{Spec}\,(\mathcal{O}_K)$ for some number field K.

Theorem (Charles)

Take two elliptic curves E, E' over K. There are infinitely many primes $\mathfrak{p} \in S$ such that E and E' becomes geometrically isogenous mod $\mathfrak{p}$.

The Case of K3 Surfaces

- When $h^{0,2}(\mathbb{V}) = 1$ (e.g., for K3 surfaces), the hypothesis of Green's theorem just amounts to saying that $\mathbb{V}$ is non-isotrivial.
- This special case was also observed by Oguiso.

Question

What happens when we replace the base S by an arithmetic curve?

- Say $S = \mathrm{Spec}\,(\mathcal{O}_K)$ for some number field K.

Theorem (Charles)

Take two elliptic curves E, E' over K. There are infinitely many primes $\mathfrak{p} \in S$ such that E and E' becomes geometrically isogenous mod $\mathfrak{p}$.

- Recall $\mathrm{NS}(E \times E') = \mathbb{Z}^2 \oplus \mathrm{Hom}(E, E')$. So the theorem says that the NL locus for $E \times E'$ (after spreading out) is dense on S.

Number Field Case

Theorem (Shankar-Shankar-Tang-Tayou + Tayou)

If X is a K3 surface over K, then for infinitely many $\mathfrak{p} \in S$, the mod $\mathfrak{p}$ reduction acquires more divisors.

Number Field Case

Theorem (Shankar-Shankar-Tang-Tayou + Tayou)

If X is a K3 surface over K, then for infinitely many $\mathfrak{p} \in S$, the mod $\mathfrak{p}$ reduction acquires more divisors.

- If $\operatorname{rank} \mathrm{NS}(X_{\bar{K}})$ is odd, then this is true for every $\mathfrak{p}$ due to the Weil and Tate conjectures!

Number Field Case

Theorem (Shankar-Shankar-Tang-Tayou + Tayou)

If X is a K3 surface over K, then for infinitely many $\mathfrak{p} \in S$, the mod $\mathfrak{p}$ reduction acquires more divisors.

- If $\operatorname{rank} \mathrm{NS}(X_{\bar{K}})$ is odd, then this is true for every $\mathfrak{p}$ due to the Weil and Tate conjectures!
- Every K3 over $\bar{\mathbb{F}}_p$ has even Picard rank, because Frobenius eigenvalues which are not roots of unity come in pairs.

Number Field Case

Theorem (Shankar-Shankar-Tang-Tayou + Tayou)

If X is a K3 surface over K, then for infinitely many $\mathfrak{p} \in S$, the mod $\mathfrak{p}$ reduction acquires more divisors.

- If $\operatorname{rank} \mathrm{NS}(X_{\bar{K}})$ is odd, then this is true for every $\mathfrak{p}$ due to the Weil and Tate conjectures!
- Every K3 over $\bar{\mathbb{F}}_p$ has even Picard rank, because Frobenius eigenvalues which are not roots of unity come in pairs.
- This observation was used to show that every K3 surface over $\mathbb{C}$ has infinitely many rational curves (Bogomolov-Hassett-Tschinkel, Li-Liedtke, Chen-Gounelas-Liedtke)!

Global Function Field Case

- What if S is a curve C over a finite field $\mathbb{F}_q$?

Global Function Field Case

- What if S is a curve C over a finite field $\mathbb{F}_q$?

Theorem (Maulik–Shankar–Tang + Tayou)

If $\mathcal{X}$ is a generically ordinary non-isotrivial family of K3 surfaces over C, then the NL locus on C is dense (i.e., infinite).

Global Function Field Case

- What if S is a curve C over a finite field $\mathbb{F}_q$?

Theorem (Maulik–Shankar–Tang + Tayou)

If $\mathcal{X}$ is a generically ordinary non-isotrivial family of K3 surfaces over C, then the NL locus on C is dense (i.e., infinite).

- Torsion of Abelian Varieties: For a dimension g abelian variety A, the torsion behavior depends on the characteristic:

$$\begin{cases} \text{Char } 0: & A[n](\overline{k}) \cong (\mathbb{Z}/n\mathbb{Z})^{2g} \text{ for all } n. \\ \text{Char } p: & A \text{ is ordinary iff } A[p](\overline{k}) \cong (\mathbb{Z}/p\mathbb{Z})^{g} \text{ (maximal } p\text{-rank).} \end{cases}$$

Global Function Field Case

- What if S is a curve C over a finite field $\mathbb{F}_q$?

Theorem (Maulik–Shankar–Tang + Tayou)

If $\mathcal{X}$ is a generically ordinary non-isotrivial family of K3 surfaces over C, then the NL locus on C is dense (i.e., infinite).

- Torsion of Abelian Varieties: For a dimension g abelian variety A, the torsion behavior depends on the characteristic:

$$\begin{cases} \text{Char } 0: & A[n](\overline{k}) \cong (\mathbb{Z}/n\mathbb{Z})^{2g} \text{ for all } n. \\ \text{Char } p: & A \text{ is ordinary iff } A[p](\overline{k}) \cong (\mathbb{Z}/p\mathbb{Z})^{g} \text{ (maximal } p\text{-rank).} \end{cases}$$

- In characteristic p, "ordinary" varieties are the least pathological cases, behaving much like their counterparts over $\mathbb{C}$.

Ordinariness: Torsion and Point Counting

- Point Counting for Elliptic Curves: For an elliptic curve E over $\mathbb{F}_q$, we count points via $\#E(\mathbb{F}_q) = q + 1 - a_q$, where a_q is the trace of Frobenius (Hasse: $|a_q| \leq 2\sqrt{q}$).

Ordinariness: Torsion and Point Counting

- Point Counting for Elliptic Curves: For an elliptic curve E over $\mathbb{F}_q$, we count points via $\#E(\mathbb{F}_q) = q + 1 - a_q$, where a_q is the trace of Frobenius (Hasse: $|a_q| \leq 2\sqrt{q}$).
- We can classify E using a simple congruence:

$$E \text{ is } \begin{cases} \text{ordinary} & \text{iff } a_q \not\equiv 0 \pmod{p} \\ \text{supersingular} & \text{iff } a_q \equiv 0 \pmod{p} \end{cases}$$

Ordinariness: Torsion and Point Counting

- Point Counting for Elliptic Curves: For an elliptic curve E over $\mathbb{F}_q$, we count points via $\#E(\mathbb{F}_q) = q + 1 - a_q$, where a_q is the trace of Frobenius (Hasse: $|a_q| \leq 2\sqrt{q}$).
- We can classify E using a simple congruence:

$$E \text{ is } \begin{cases} \text{ordinary} & \text{iff } a_q \not\equiv 0 \pmod{p} \\ \text{supersingular} & \text{iff } a_q \equiv 0 \pmod{p} \end{cases}$$

- For K3 surfaces over a char p field k, let h be the height of the formal Brauer group.

Ordinariness: Torsion and Point Counting

- Point Counting for Elliptic Curves: For an elliptic curve E over $\mathbb{F}_q$, we count points via $\#E(\mathbb{F}_q) = q + 1 - a_q$, where a_q is the trace of Frobenius (Hasse: $|a_q| \leq 2\sqrt{q}$).
- We can classify E using a simple congruence:

$$E \text{ is } \begin{cases} \text{ordinary} & \text{iff } a_q \not\equiv 0 \pmod{p} \\ \text{supersingular} & \text{iff } a_q \equiv 0 \pmod{p} \end{cases}$$

- For K3 surfaces over a char p field k, let h be the height of the formal Brauer group.
- If $h < \infty$, it is also the number of Frobenius eigenvalues of p-adic valuation < 1. If $h = \infty$ (supersingular), then all Frobenius eigenvalues have p-adic valuation 1.

Beyond Ordinariness: A Counterexample

- But what if the family $\mathcal{X} \to S$ is not generically ordinary?

Beyond Ordinariness: A Counterexample

- But what if the family $\mathcal{X} \to S$ is not generically ordinary?
- Maulik-Shankar-Tang: Consider an abelian surface family over an arithmetic curve S:

$$\mathcal{X} = \mathcal{E} \times \mathcal{E}' \quad \text{where} \quad \begin{cases} \mathcal{E}' = E' \times S & \text{(constant supersingular)} \\ \mathcal{E} \to S & \text{(non-isotrivial)} \end{cases}$$

Beyond Ordinariness: A Counterexample

- But what if the family $\mathcal{X} \to S$ is not generically ordinary?
- Maulik-Shankar-Tang: Consider an abelian surface family over an arithmetic curve S:

$$\mathcal{X} = \mathcal{E} \times \mathcal{E}' \quad \text{where} \quad \begin{cases} \mathcal{E}' = E' \times S & \text{(constant supersingular)} \\ \mathcal{E} \to S & \text{(non-isotrivial)} \end{cases}$$

- The Picard rank of the fiber $\mathcal{X}_s = \mathcal{E}_s \times E'$ jumps at s when $\mathcal{E}_s$ becomes isogenous to E'.

Beyond Ordinariness: A Counterexample

- But what if the family $\mathcal{X} \to S$ is not generically ordinary?
- Maulik-Shankar-Tang: Consider an abelian surface family over an arithmetic curve S:

$$\mathcal{X} = \mathcal{E} \times \mathcal{E}' \quad \text{where} \quad \begin{cases} \mathcal{E}' = E' \times S & \text{(constant supersingular)} \\ \mathcal{E} \to S & \text{(non-isotrivial)} \end{cases}$$

- The Picard rank of the fiber $\mathcal{X}_s = \mathcal{E}_s \times E'$ jumps at s when $\mathcal{E}_s$ becomes isogenous to E'.
- This is equivalent to asking the fiber $\mathcal{E}_s$ to be supersingular.

Beyond Ordinariness: A Counterexample

- But what if the family $\mathcal{X} \to S$ is not generically ordinary?
- Maulik-Shankar-Tang: Consider an abelian surface family over an arithmetic curve S:

$$\mathcal{X} = \mathcal{E} \times \mathcal{E}' \quad \text{where} \quad \begin{cases} \mathcal{E}' = E' \times S & \text{(constant supersingular)} \\ \mathcal{E} \to S & \text{(non-isotrivial)} \end{cases}$$

- The Picard rank of the fiber $\mathcal{X}_s = \mathcal{E}_s \times E'$ jumps at s when $\mathcal{E}_s$ becomes isogenous to E'.
- This is equivalent to asking the fiber $\mathcal{E}_s$ to be supersingular.
- Because a non-isotrivial family of elliptic curves has only finitely many supersingular fibers, the Noether-Lefschetz locus for $\mathcal{X}$ is finite and strictly non-dense!

Our Main Theorem

- Goal: Understand exactly what is going on.

Our Main Theorem

- Goal: Understand exactly what is going on.
- Problem: The above example is "too close to being supersingular".

Our Main Theorem

- Goal: Understand exactly what is going on.
- Problem: The above example is "too close to being supersingular".
- Supersingular: whatever may go wrong in char p goes wrong. It is the antithesis of being ordinary.

Our Main Theorem

- Goal: Understand exactly what is going on.
- Problem: The above example is "too close to being supersingular".
- Supersingular: whatever may go wrong in char p goes wrong. It is the antithesis of being ordinary.
- A supersingular K3 has Picard rank 22 = second Betti number. It may even be unirational!

Our Main Theorem

- Goal: Understand exactly what is going on.
- Problem: The above example is "too close to being supersingular".
- Supersingular: whatever may go wrong in char p goes wrong. It is the antithesis of being ordinary.
- A supersingular K3 has Picard rank 22 = second Betti number. It may even be unirational!
- Let $\mathscr{S} := \mathscr{S}(L)$ be the smallest orthogonal Shimura variety that contains C. Here L is a suitable lattice of signature $(2, b)$.

Our Main Theorem

- Goal: Understand exactly what is going on.
- Problem: The above example is "too close to being supersingular".
- Supersingular: whatever may go wrong in char p goes wrong. It is the antithesis of being ordinary.
- A supersingular K3 has Picard rank 22 = second Betti number. It may even be unirational!
- Let $\mathscr{S} := \mathscr{S}(L)$ be the smallest orthogonal Shimura variety that contains C. Here L is a suitable lattice of signature $(2, b)$.
- The mod p fiber $\mathscr{S}_{\mathbb{F}_p}$ has a Newton (or height) stratification.

Our Main Theorem

- Goal: Understand exactly what is going on.
- Problem: The above example is "too close to being supersingular".
- Supersingular: whatever may go wrong in char p goes wrong. It is the antithesis of being ordinary.
- A supersingular K3 has Picard rank 22 = second Betti number. It may even be unirational!
- Let $\mathscr{S} := \mathscr{S}(L)$ be the smallest orthogonal Shimura variety that contains C. Here L is a suitable lattice of signature $(2, b)$.
- The mod p fiber $\mathscr{S}_{\mathbb{F}_p}$ has a Newton (or height) stratification.
- We say that $\mathcal{X} \to C$ is exceptional if

$$\begin{cases} \text{it is supersingular, or} \\ \text{it is almost supersingular and } L \otimes \mathbb{Z}_p \simeq (U \otimes \mathbb{Z}_p)^{\oplus(b/2+1)}. \end{cases}$$

Our Main Theorem

Theorem (Jiang-Shankar-Y)

If a non-isotrivial family of K3 surfaces $\mathcal{X} \to C$ is not exceptional, and $(*)$ the connected-étale exact sequence of the enlarged formal Brauer group of the generic fiber does not split, then the NL locus is infinite.

Our Main Theorem

Theorem (Jiang-Shankar-Y)

If a non-isotrivial family of K3 surfaces $\mathcal{X} \to C$ is not exceptional, and $(*)$ the connected-étale exact sequence of the enlarged formal Brauer group of the generic fiber does not split, then the NL locus is infinite.

- "not exceptional" is excluding an obvious obstruction from Frobenius for the statement to be true.

Our Main Theorem

Theorem (Jiang-Shankar-Y)

If a non-isotrivial family of K3 surfaces $\mathcal{X} \to C$ is not exceptional, and $(*)$ the connected-étale exact sequence of the enlarged formal Brauer group of the generic fiber does not split, then the NL locus is infinite.

- "not exceptional" is excluding an obvious obstruction from Frobenius for the statement to be true.
- The previous counterexample is exceptional, but there are also non-exceptional countexamples.

Our Main Theorem

Theorem (Jiang-Shankar-Y)

If a non-isotrivial family of K3 surfaces $\mathcal{X} \to C$ is not exceptional, and $(*)$ the connected-étale exact sequence of the enlarged formal Brauer group of the generic fiber does not split, then the NL locus is infinite.

- "not exceptional" is excluding an obvious obstruction from Frobenius for the statement to be true.
- The previous counterexample is exceptional, but there are also non-exceptional countexamples.
- $(*)$ should be viewed as saying that the crystalline monodromy group of the family is not too degenerate.

Our Main Theorem

Theorem (Jiang-Shankar-Y)

If a non-isotrivial family of K3 surfaces $\mathcal{X} \to C$ is not exceptional, and $(*)$ the connected-étale exact sequence of the enlarged formal Brauer group of the generic fiber does not split, then the NL locus is infinite.

- "not exceptional" is excluding an obvious obstruction from Frobenius for the statement to be true.
- The previous counterexample is exceptional, but there are also non-exceptional countexamples.
- $(*)$ should be viewed as saying that the crystalline monodromy group of the family is not too degenerate.
- $(*)$ is automatically satisfied if the family is generically ordinary.

Proof Strategy

- The Core Strategy: Intersect our arithmetic curve C with a family of special divisors $\mathcal{Z}(m)$ on the ambient Shimura variety.

Proof Strategy

- The Core Strategy: Intersect our arithmetic curve C with a family of special divisors $\mathcal{Z}(m)$ on the ambient Shimura variety.
- What is $\mathcal{Z}(m)$? It is the special divisor parameterizing points where the fibers acquire an additional divisor of degree m.

Proof Strategy

- The Core Strategy: Intersect our arithmetic curve C with a family of special divisors $\mathcal{Z}(m)$ on the ambient Shimura variety.
- What is $\mathcal{Z}(m)$? It is the special divisor parameterizing points where the fibers acquire an additional divisor of degree m.
- The generating series for these special divisors is modular. Using Borcherds's theory, we obtain that $C \cdot \mathcal{Z}(m) \asymp m^{(b+2)/2}$.

Proof Strategy

- The Core Strategy: Intersect our arithmetic curve C with a family of special divisors $\mathcal{Z}(m)$ on the ambient Shimura variety.
- What is $\mathcal{Z}(m)$? It is the special divisor parameterizing points where the fibers acquire an additional divisor of degree m.
- The generating series for these special divisors is modular. Using Borcherds's theory, we obtain that $C \cdot \mathcal{Z}(m) \asymp m^{(b+2)/2}$.
- Now given P, we bound the growth rate of the intersection multiplicity $i_P(C \cdot \mathcal{Z}(m))$ at P by deformation theory.

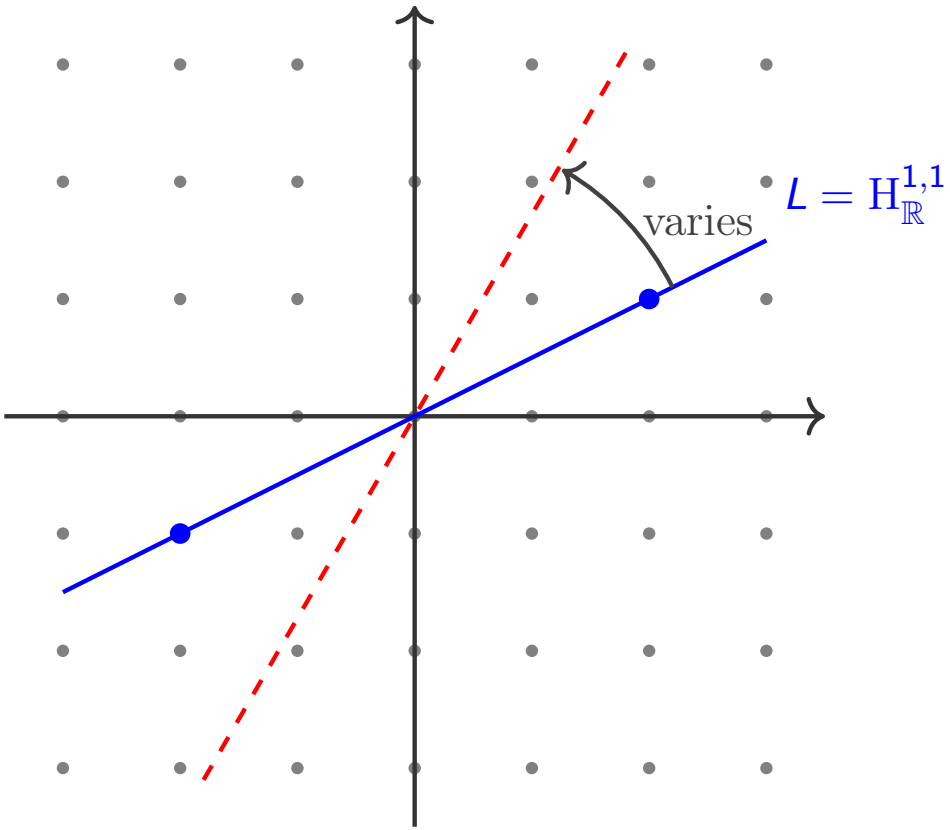
Proof Strategy

- The Core Strategy: Intersect our arithmetic curve C with a family of special divisors $\mathcal{Z}(m)$ on the ambient Shimura variety.
- What is $\mathcal{Z}(m)$? It is the special divisor parameterizing points where the fibers acquire an additional divisor of degree m.
- The generating series for these special divisors is modular. Using Borcherds's theory, we obtain that $C \cdot \mathcal{Z}(m) \asymp m^{(b+2)/2}$.
- Now given P, we bound the growth rate of the intersection multiplicity $i_P(C \cdot \mathcal{Z}(m))$ at P by deformation theory.
- When P is supersingular, $i_P(C \cdot \mathcal{Z}(m)) \asymp m^{(b+2)/2}$, so one needs delicate control on the constants appearing in "$\asymp$".

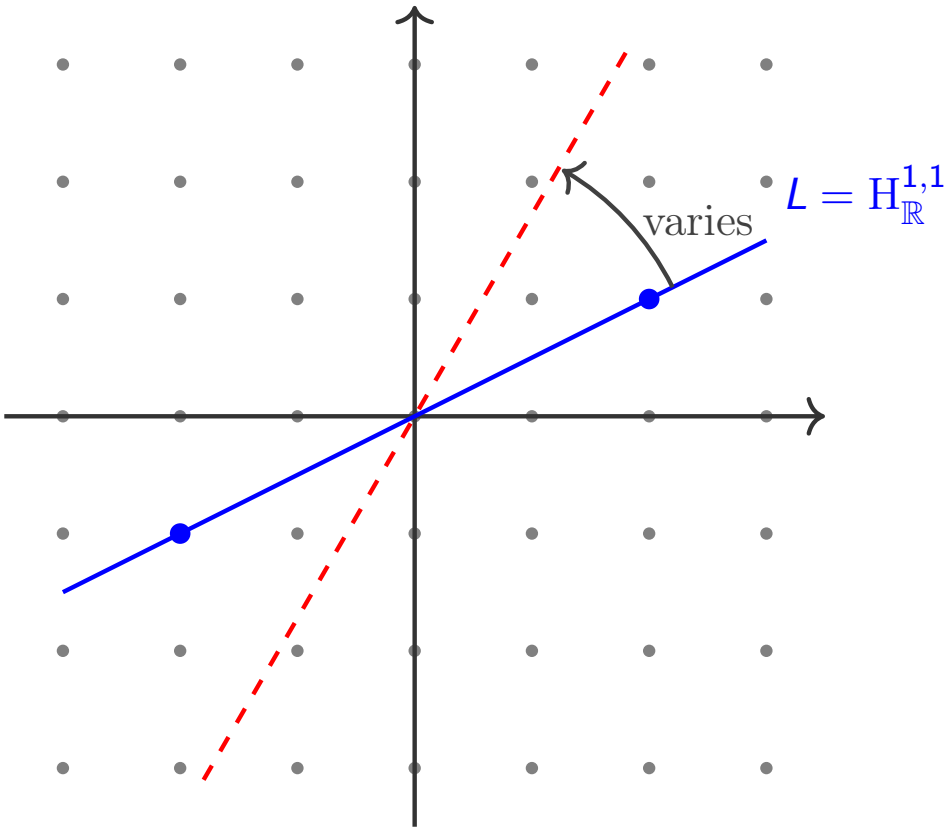
Proof Strategy

- The Core Strategy: Intersect our arithmetic curve C with a family of special divisors $\mathcal{Z}(m)$ on the ambient Shimura variety.
- What is $\mathcal{Z}(m)$? It is the special divisor parameterizing points where the fibers acquire an additional divisor of degree m.
- The generating series for these special divisors is modular. Using Borcherds's theory, we obtain that $C \cdot \mathcal{Z}(m) \asymp m^{(b+2)/2}$.
- Now given P, we bound the growth rate of the intersection multiplicity $i_P(C \cdot \mathcal{Z}(m))$ at P by deformation theory.
- When P is supersingular, $i_P(C \cdot \mathcal{Z}(m)) \asymp m^{(b+2)/2}$, so one needs delicate control on the constants appearing in "$\asymp$".
- This step gets much more delicate in the non-ordinary case and the tangency of C with each Newton stratum plays a centrol role.

Comparison with $\mathbb{C}$

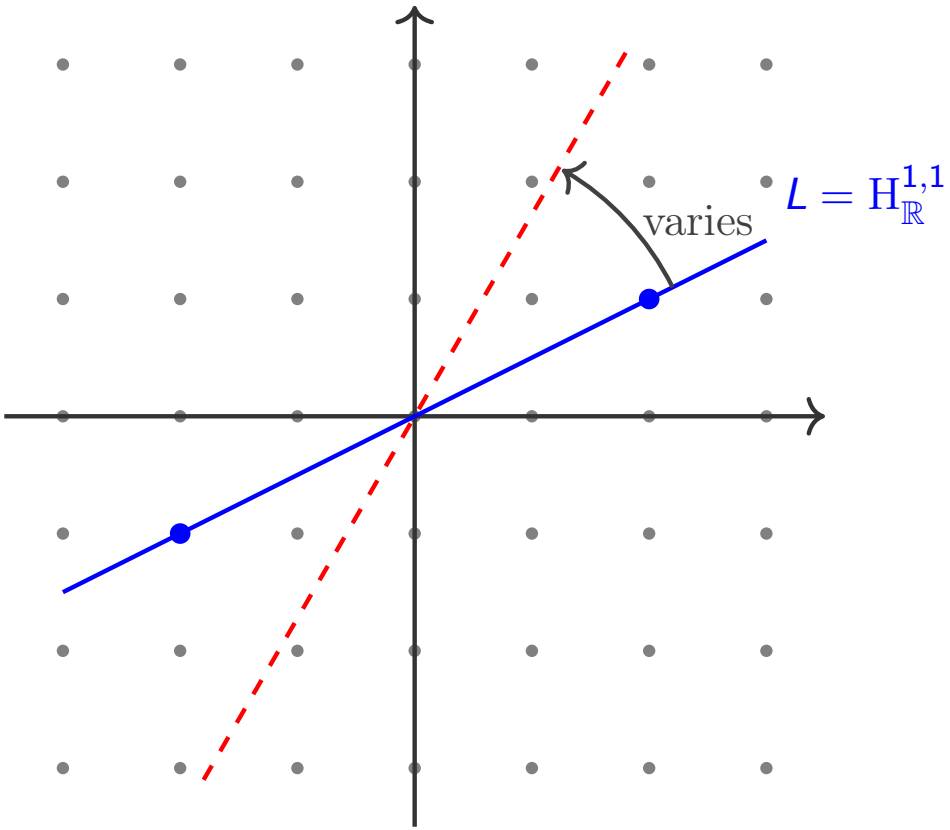


Comparison with $\mathbb{C}$



- The char p argument looks nothing like the char 0 argument.

Comparison with $\mathbb{C}$



- The char p argument looks nothing like the char 0 argument.
- But there is still some parallel with the situation over $\mathbb{C}$.

- There is a lot of recent progress on the following (Klingler, Otwinowska, Baldi, Ullmo, Urbanik, Khelifa, Tayou, Tholozan...)

- There is a lot of recent progress on the following (Klingler, Otwinowska, Baldi, Ullmo, Urbanik, Khelifa, Tayou, Tholozan...)

Question

When is the Hodge locus, or a certain part of it, dense?

- There is a lot of recent progress on the following (Klingler, Otwinowska, Baldi, Ullmo, Urbanik, Khelifa, Tayou, Tholozan...)

Question

When is the Hodge locus, or a certain part of it, dense?

- In general, to make sure that a certain part of Hodge locus is dense, one needs two types of conditions:

$$\begin{cases} \text{the expected dimension of the locus is } \geq 0, \text{ and} \\ \text{the monodromy group is “not too degenerate”.} \end{cases}$$

- There is a lot of recent progress on the following (Klingler, Otwinowska, Baldi, Ullmo, Urbanik, Khelifa, Tayou, Tholozan...)

Question

When is the Hodge locus, or a certain part of it, dense?

- In general, to make sure that a certain part of Hodge locus is dense, one needs two types of conditions:

$$\begin{cases} \text{the expected dimension of the locus is } \geq 0, \text{ and} \\ \text{the monodromy group is “not too degenerate”.} \end{cases}$$

- For a weight 2 VHS $\mathbb{V}$ with $h^{2,0} = 1$, the two coincide.

- There is a lot of recent progress on the following (Klingler, Otwinowska, Baldi, Ullmo, Urbanik, Khelifa, Tayou, Tholozan...)

Question

When is the Hodge locus, or a certain part of it, dense?

- In general, to make sure that a certain part of Hodge locus is dense, one needs two types of conditions:

$$\begin{cases} \text{the expected dimension of the locus is } \geq 0, \text{ and} \\ \text{the monodromy group is “not too degenerate”.} \end{cases}$$

- For a weight 2 VHS $\mathbb{V}$ with $h^{2,0} = 1$, the two coincide.
- But they depart in char p: the supersingular locus has trivial (finite) monodromy but nonzero Kodaira-Spencer map.

Concluding remarks

- General theme: We wish to understand the distribution of divisors, or more generally, algebraic cycles in a family.

Concluding remarks

- General theme: We wish to understand the distribution of divisors, or more generally, algebraic cycles in a family.
- Over $\mathbb{C}$, a lot is known by studying variation of Hodge structures.

Concluding remarks

- General theme: We wish to understand the distribution of divisors, or more generally, algebraic cycles in a family.
- Over $\mathbb{C}$, a lot is known by studying variation of Hodge structures.
- In char p, we believe that ordinary varieties should behave like the case over $\mathbb{C}$ (at least in a Shimura setting, cf. AIM conjecture).

Concluding remarks

- General theme: We wish to understand the distribution of divisors, or more generally, algebraic cycles in a family.
- Over $\mathbb{C}$, a lot is known by studying variation of Hodge structures.
- In char p, we believe that ordinary varieties should behave like the case over $\mathbb{C}$ (at least in a Shimura setting, cf. AIM conjecture).
- But for non-ordinary case, even less is known, but we do want to understand exactly how the Frobenius constrains the distribution of cycles, or in a Shimura setting, Hecke orbits.

Concluding remarks

- General theme: We wish to understand the distribution of divisors, or more generally, algebraic cycles in a family.
- Over $\mathbb{C}$, a lot is known by studying variation of Hodge structures.
- In char p, we believe that ordinary varieties should behave like the case over $\mathbb{C}$ (at least in a Shimura setting, cf. AIM conjecture).
- But for non-ordinary case, even less is known, but we do want to understand exactly how the Frobenius constrains the distribution of cycles, or in a Shimura setting, Hecke orbits.
- Related works: D'Addezio and van Hoften's proof of the full Chai-Oort Hecke orbit conjecture, Jiang's work on almost ordinary abelian surfaces.

Concluding remarks

- General theme: We wish to understand the distribution of divisors, or more generally, algebraic cycles in a family.
- Over $\mathbb{C}$, a lot is known by studying variation of Hodge structures.
- In char p, we believe that ordinary varieties should behave like the case over $\mathbb{C}$ (at least in a Shimura setting, cf. AIM conjecture).
- But for non-ordinary case, even less is known, but we do want to understand exactly how the Frobenius constrains the distribution of cycles, or in a Shimura setting, Hecke orbits.
- Related works: D'Addezio and van Hoften's proof of the full Chai-Oort Hecke orbit conjecture, Jiang's work on almost ordinary abelian surfaces.
- Thank you!